Modelling the Volatility of the FTSE All Share Index: Evidence from GARCH and TARCH Models

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Abstract

This paper investigates the volatility dynamics of the FTSE All Share Index using daily financial data from January 2010 to March 2023. The analysis focuses on whether the return series exhibits volatility clustering, time-varying conditional variance, and asymmetric responses to positive and negative shocks. After documenting severe non-normality and strong ARCH effects in the preliminary data, we estimate GARCH(1,1) and TARCH/GJR-GARCH(1,1) models utilizing Student-t innovations to appropriately account for fat tails. The empirical results confirm a high degree of volatility persistence in the UK equity market. Furthermore, the GJR-GARCH specification reveals a statistically significant leverage effect, indicating that negative return shocks increase future volatility more severely than positive shocks of an equal magnitude. Post-estimation diagnostics confirm that the asymmetric model adequately captures the volatility clustering present in the data.

Keywords: FTSE All Share Index, volatility, GARCH, TARCH, financial returns, conditional heteroskedasticity

1 Introduction

Volatility is one of the central concepts in financial econometrics, because it represents the degree of uncertainty and risk associated with asset returns. For investors, portfolio managers, and policymakers, understanding volatility is important for risk management, asset allocation, derivative pricing, and the assessment of financial market stability. Unlike the level of asset returns, which is often difficult to predict, financial volatility tends to display systematic patterns over time. In particular, financial return series commonly exhibit volatility clustering, meaning that periods of high volatility are often followed by further periods of high volatility, while tranquil periods tend to persist as well.

This paper studies the volatility dynamics of the FTSE All Share Index, a broad stock market index representing the performance of companies listed on the London Stock Exchange. Since the index covers a wide segment of the UK equity market, its volatility can be interpreted as an indicator of uncertainty in the broader UK stock market. Modelling this volatility is therefore relevant not only from the perspective of investors, but also for understanding how financial risk evolves over time in a major equity market.

The empirical analysis focuses on daily returns rather than price levels. This is because financial price series are often non-stationary, while return series are typically more suitable for time-series modelling. After transforming the FTSE All Share Index prices into continuously compounded daily returns, we examine their statistical properties and test whether the series exhibits features commonly associated with financial data, such as fat tails, volatility clustering, and time-varying conditional variance.

To model conditional volatility, we use the ARCH framework introduced by Engle (1982) and the GARCH model generalized by Bollerslev (1986). The baseline GARCH model allows current volatility to depend on both past shocks and past volatility, making it suitable for capturing volatility persistence. However, standard GARCH models treat positive and negative shocks symmetrically. In financial markets, negative shocks may have a stronger effect on future volatility than positive shocks of the same magnitude. Therefore, we also estimate a TARCH-type model, closely related to the asymmetric GJR-GARCH specification of Glosten et al. (1993) and threshold heteroskedastic models discussed by Zakoian (1994). As an optional robustness extension, we may also consider the EGARCH model proposed by Nelson (1991).

The main research question of this paper is whether the FTSE All Share Index exhibits volatility clustering and asymmetric volatility effects that can be captured by GARCH and TARCH models. More specifically, we investigate whether volatility is persistent over time and whether negative return shocks increase future volatility more strongly than positive shocks. Similar GARCH-type approaches have been widely used in empirical studies of financial volatility, including applications to exchange rate volatility such as Fidrmuc and Horvath (2008).

The remainder of the paper is structured as follows. Section 2 describes the data and the transformation of prices into returns. Section 3 presents preliminary time-series analysis, including descriptive statistics, stationarity tests, and autocorrelation diagnostics. Section 4 introduces the econometric methodology, focusing on GARCH and TARCH models. Section 5 reports and interprets the empirical results. Section 6 discusses model diagnostics, and Section 7 concludes.

2 Data Description

The dataset consists of daily closing prices of the FTSE All Share Index. The original dataset contains 3,328 observations. One missing value is present in the raw closing-price series and is removed before the empirical analysis. After cleaning the data, the price series contains 3,327 daily observations from 4 January 2010 to 6 March 2023.

The raw price series is transformed into continuously compounded daily returns using:

$$r_t = 100 \times (\log(P_t) - \log(P_{t-1})), \tag{1}$$

where P_t is the closing value of the index on day t and r_t is the daily log return in percentage terms. After the transformation, the return series contains 3,326 observations.

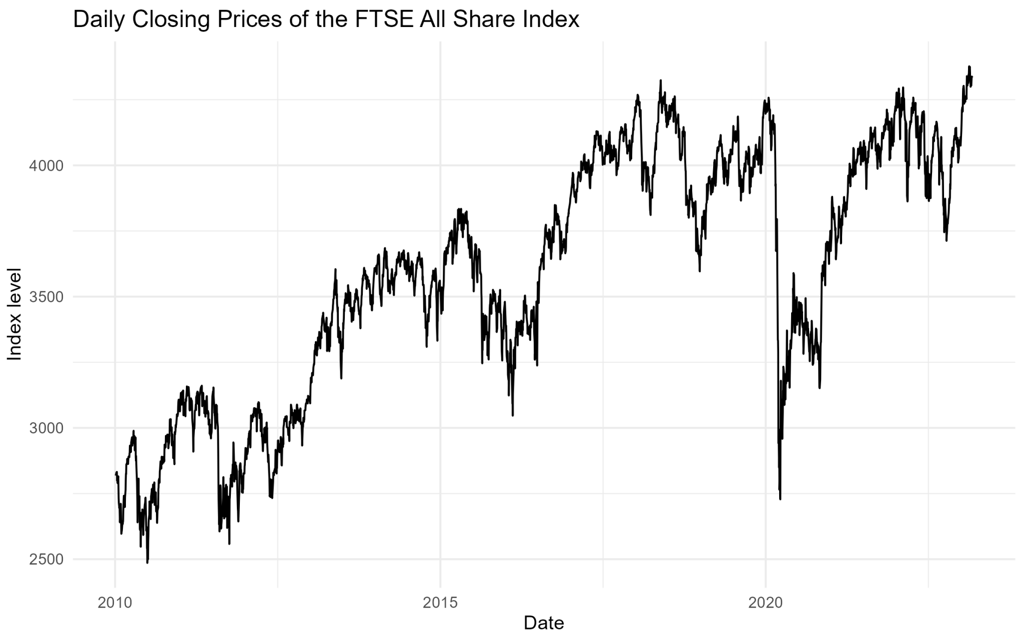

Figure 1: Daily closing prices of the FTSE All Share Index

Figure 1 shows the daily closing prices of the FTSE All Share Index. The index level displays substantial variation over time, including a sharp decline around 2020 followed by a recovery. The visual pattern suggests that the price level is not stable around a constant mean, which motivates the use of returns rather than raw prices in the empirical analysis.

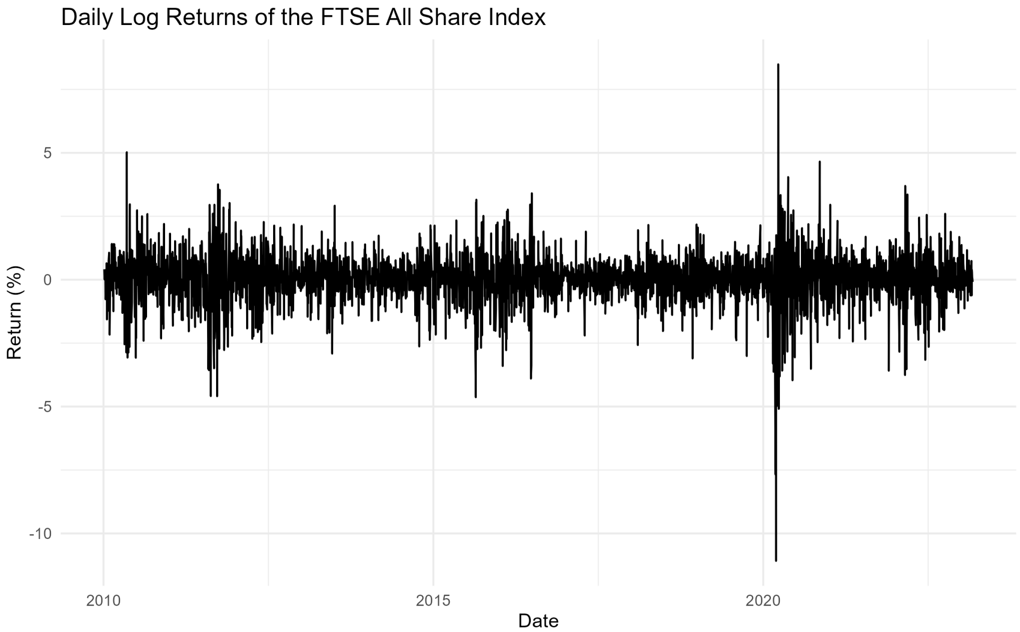

Figure 2: Daily log returns of the FTSE All Share Index

Figure 2 plots the daily log returns of the FTSE All Share Index. The return series fluctuates around zero, but the magnitude of fluctuations changes over time. Periods of relatively calm movements alternate with periods of turbulence, especially around the COVID-19 crisis. This visual pattern provides initial evidence of volatility clustering.

Table 1: Descriptive statistics of FTSE All Share returns

Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	Observations
0.0131	0.9945	-11.0836	8.4903	-0.6777	12.6057	3326

Table 1 summarizes the basic distributional properties of the return series. The average daily return is positive but very small, while the standard deviation is close to one percent. The minimum daily return is substantially larger in absolute value than the maximum positive return, suggesting a stronger negative tail. The skewness is negative and the kurtosis is far above the value of three associated with a normal distribution. The Jarque-Bera test strongly rejects the null hypothesis of normality, with a test statistic of 13042 and a p-value below 2.2×10^{-16}. These results indicate that the return series displays typical features of financial returns, including non-normality and fat tails.

3 Preliminary Time-Series Analysis

This section examines whether the return series is suitable for GARCH-type modelling. We first test the stationarity of the price and return series. We then examine the autocorrelation structure of returns and squared returns using autocorrelation functions and Ljung-Box tests. Finally, we estimate simple mean equations and test their residuals for ARCH effects.

3.1 Stationarity Tests

Table 2: Stationarity tests for prices and returns

Series	Test	Test Statistic	p-value
Price level	ADF	-3.5422	0.0382
Returns	ADF	-15.1540	0.0100
Price level	KPSS	22.6550	0.0100
Returns	KPSS	0.0220	0.1000

Table 2 reports the ADF and KPSS stationarity tests. The KPSS test rejects level stationarity for the price series, while it does not reject stationarity for the return series. The ADF test strongly rejects the presence of a unit root in returns. The ADF test also rejects a unit root in the price level at the five percent level, while the KPSS test rejects stationarity of the price level. Because the two tests provide mixed evidence for the price level, we avoid relying on the level series and focus on returns. Overall, the return series is clearly more suitable for time-series modelling.

3.2 Autocorrelation and Volatility Clustering

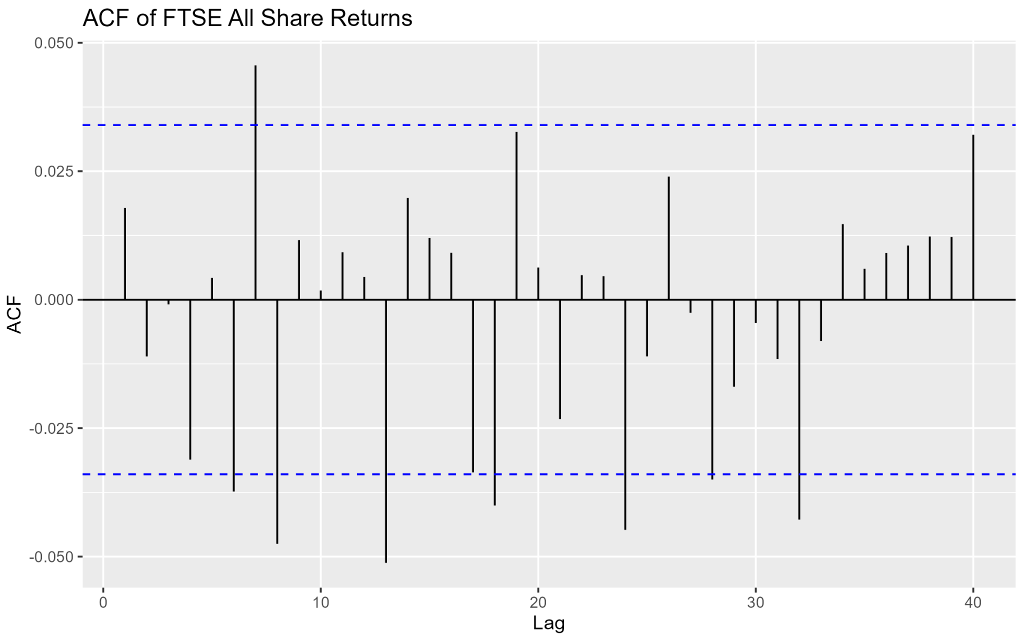

Figure 3: Autocorrelation function of FTSE All Share returns

Figure 3 shows the autocorrelation function of FTSE All Share returns. The autocorrelations of raw returns are relatively small, although some lags cross the approximate significance bands. This suggests that the return series contains some serial dependence, but the dependence in the conditional mean is not the main feature of the data.

Figure 4: Autocorrelation function of squared FTSE All Share returns

Figure 4 shows the autocorrelation function of squared returns. The autocorrelations are positive and persistent across many lags. This is strong visual evidence of volatility clustering: large shocks tend to be followed by large shocks, while small shocks tend to be followed by small shocks.

Table 3: Ljung-Box tests for returns and squared returns

Series	Lag	Test Statistic	p-value
Returns	10	24.3240	0.0068
Returns	20	48.3540	0.0004
Squared returns	10	1541.0000	$< 2.2 \times 10^{-16}$
Squared returns	20	2072.1000	$< 2.2 \times 10^{-16}$

The Ljung-Box results in Table 3 confirm the graphical evidence. The test rejects the null hypothesis of no autocorrelation for both returns and squared returns. The evidence is especially strong for squared returns, where the p-values are effectively zero. This suggests the presence of time-varying volatility and motivates the use of ARCH and GARCH-type models.

3.3 Mean Equation and ARCH Effects

Before estimating volatility models, we consider simple specifications for the conditional mean equation. Using the AIC, the selected model is an ARIMA(2,0,2) model with zero mean. Using the BIC, the selected model is an ARIMA(0,0,0) model with zero mean. This difference is expected, since AIC often favours more flexible models, while BIC imposes a stronger penalty for additional parameters.

Table 4: Mean model comparison

Mean model	Log-likelihood	AIC	BIC
ARIMA(2,0,2), zero mean	-4692.21	9394.42	9424.97
ARIMA(0,0,0), zero mean	-4700.86	9403.71	9409.82

Table 4 shows that the ARIMA(2,0,2) model has a lower AIC, while the zero-mean model has a lower BIC. Since the main focus of the paper is volatility rather than return predictability, the mean equation is treated mainly as a preliminary step before modelling conditional variance.

Table 5: ARCH-LM tests for residuals from mean equations

Mean model	Lags	Test Statistic	p-value
ARIMA(2,0,2), zero mean	5	570.42	$< 2.2 \times 10^{-16}$
ARIMA(2,0,2), zero mean	10	724.49	$< 2.2 \times 10^{-16}$
ARIMA(2,0,2), zero mean	20	783.50	$< 2.2 \times 10^{-16}$
ARIMA(0,0,0), zero mean	5	518.75	$< 2.2 \times 10^{-16}$
ARIMA(0,0,0), zero mean	10	716.58	$< 2.2 \times 10^{-16}$
ARIMA(0,0,0), zero mean	20	780.76	$< 2.2 \times 10^{-16}$

Table 5 reports ARCH-LM tests applied to residuals from both mean equations. In all cases, the null hypothesis of no ARCH effects is strongly rejected. Therefore, regardless of the chosen mean equation, the residuals exhibit conditional heteroskedasticity. This provides strong justification for estimating GARCH and TARCH models in the next step.

4 Empirical Methodology

This section explains the econometric models used in the paper.

We first specify a mean equation for returns. A simple version is:

$$r_t = \mu + \varepsilon_t, \tag{2}$$

where r_t denotes the daily return and ε_t is the error term.

If necessary, the mean equation may include autoregressive or moving-average terms, depending on the autocorrelation diagnostics. This step is important because GARCH-type models are applied to the residuals from the conditional mean equation.

4.1 ARCH Effects

Before estimating GARCH models, we test whether the residuals from the mean equation exhibit ARCH effects. Evidence of ARCH effects means that the conditional variance is not constant over time and that GARCH-type models are appropriate. In practice, we use tests such as the ARCH-LM test and autocorrelation diagnostics for squared residuals.

4.2 GARCH Model

The baseline volatility model is the GARCH(1,1) model:

$$h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}, \tag{3}$$

where h_t is the conditional variance, ω is the constant component, α measures the reaction of volatility to new shocks, and β measures volatility persistence.

This model captures volatility clustering by allowing current volatility to depend on past shocks and past volatility. A high value of $\alpha + \beta$ indicates that volatility shocks are persistent and decay slowly over time.

4.3 TARCH / GJR-GARCH Model

To examine asymmetric volatility effects, we estimate a threshold-type GARCH model. The model allows negative shocks to have a different effect on volatility than positive shocks. A common specification is the GJR-GARCH(1,1) model:

$$h_t = \omega + \alpha\varepsilon_{t-1}^2 + \gamma I_{\{\varepsilon_{t-1}<0\}}\varepsilon_{t-1}^2 + \beta h_{t-1}, \tag{4}$$

where $I_{\{\varepsilon_{t-1}<0\}}$ is an indicator variable equal to one when the previous shock is negative and zero otherwise. The coefficient γ measures the additional effect of negative shocks on future volatility. If γ is positive and statistically significant, negative shocks increase volatility more strongly than positive shocks of the same size.

The general idea is that bad news may increase volatility more strongly than good news. This is often called the leverage effect or asymmetric volatility effect.

5 Empirical Results

To model the conditional volatility of the FTSE All Share Index returns, we proceed with estimating the GARCH(1,1) and GJR-GARCH(1,1) models. Given the preliminary finding that the return series is highly leptokurtic and the Jarque-Bera test strongly rejects normality, we estimate all models using Student-t distributed innovations. Furthermore, because predicting the exact daily return is not the primary focus of this analysis, we utilize a simple zero-mean equation, which is supported by the BIC from our preliminary selection.

5.1 Baseline GARCH(1,1) Results

Table 6 presents the estimation results for the standard GARCH(1,1) model.

Table 6: GARCH(1,1) estimation results

Parameter	Estimate	Robust Standard Error	p-value
ω	0.0307	0.0096	0.0013
α	0.1331	0.0250	0.0000
β	0.8384	0.0309	0.0000
Shape (Student-t)	6.1844	0.6115	0.0000

The coefficient α, which measures the short-term reaction of volatility to market shocks, is estimated at 0.1331 and is highly statistically significant. The coefficient β, representing the persistence of volatility over time, is estimated at 0.8384 and is also significant. The sum of the ARCH and GARCH terms ($\alpha + \beta$) is approximately 0.9715, which is close to one. This indicates a high degree of volatility persistence, confirming our earlier graphical intuition: once a volatility shock hits the UK equity market, it takes a considerable amount of time to die out.

5.2 TARCH / GJR-GARCH(1,1) Results

While the standard GARCH captures persistence, it forces a symmetric response to shocks. Table 7 reports the results of the GJR-GARCH(1,1) model, which tests for the leverage effect.

Table 7: TARCH/GJR-GARCH estimation results

Parameter	Estimate	Robust Standard Error	p-value
ω	0.0336	0.0083	0.0000
α	0.0000	0.0103	1.0000
γ (Asymmetry)	0.2417	0.0377	0.0000
β	0.8449	0.0268	0.0000

The asymmetry parameter, γ, is estimated at 0.2417 and is statistically significant at the 1% level. Because γ is positive and significant, it confirms the presence of a leverage effect in the FTSE All Share Index. Specifically, bad news (negative returns) generates a larger increase in future volatility than good news (positive returns) of the exact same size.

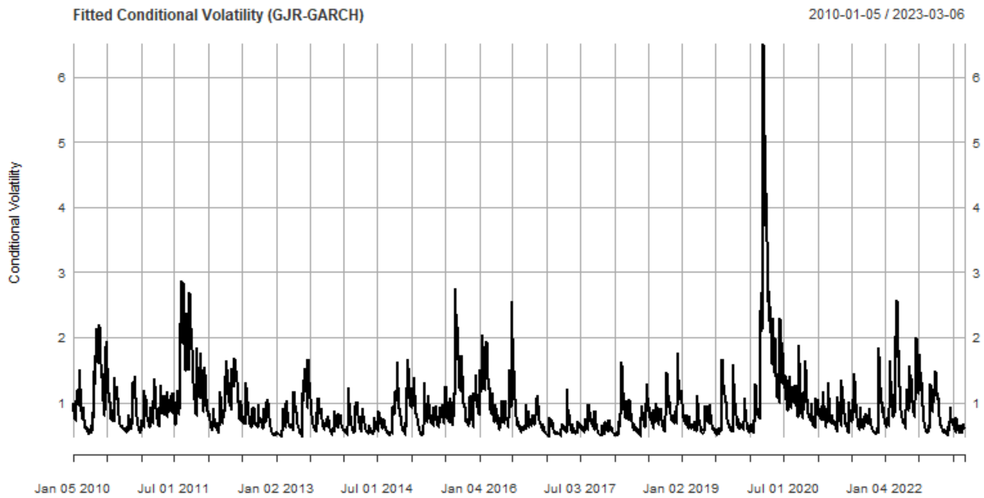

Figure 5: Fitted conditional volatility from the GJR-GARCH model

Figure 5 plots the fitted conditional volatility from the preferred GJR-GARCH model. The spikes in volatility correspond to significant market events, most notably the COVID-19 pandemic period, where volatility reached its highest levels within the sample.

5.3 Model Comparison

To objectively determine the best fit for our data, we compare the models using information criteria and log-likelihood values.

Table 8: Model comparison

Model	Log-likelihood	AIC	BIC
GARCH(1,1)	-4121.1	2.4814	2.4888
TARCH/GJR-GARCH(1,1)	-4056.1	2.4424	2.4516
EGARCH(1,1) (Appendix A)	-4047.6	2.4373	2.4465

As seen in Table 8, the GJR-GARCH model yields lower values for both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) compared to the symmetric GARCH(1,1), while increasing the log-likelihood. Consequently, the GJR-GARCH is selected as our preferred specification for interpreting the main results.

6 Model Diagnostics

A crucial step in financial econometrics is verifying whether the selected model successfully accounts for the dependence identified in the preliminary analysis. We apply diagnostic checks to the standardized residuals from the preferred GJR-GARCH model.

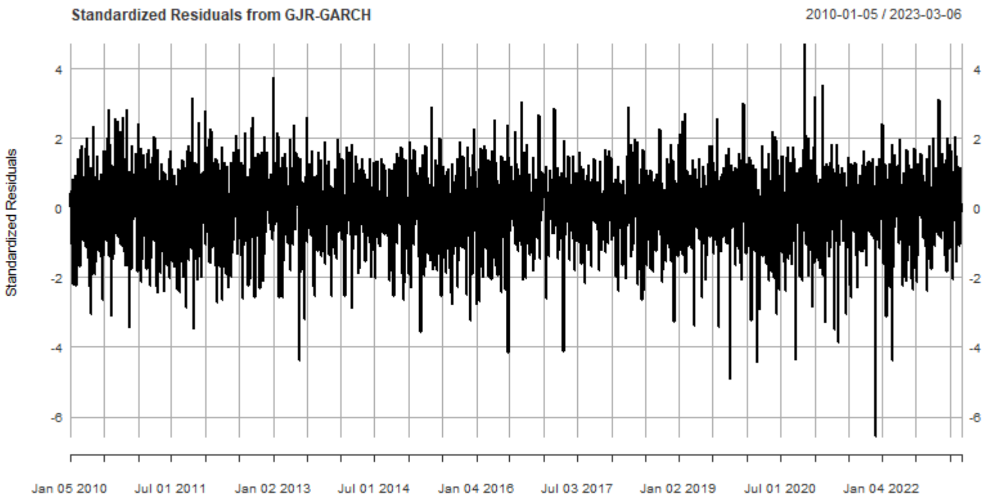

Figure 6: Standardized residuals from the GJR-GARCH model

Visually, the standardized residuals in Figure 6 show a much more stable variance profile compared to the raw returns in Figure 2, suggesting the conditional volatility has been successfully filtered.

Table 9: Diagnostic tests for standardized residuals

Test	Null Hypothesis	p-value
Ljung-Box (Lag 9)	No autocorrelation in residuals	0.1908
Ljung-Box (Lag 9)	No autocorrelation in squared residuals	0.5734
Weighted ARCH-LM (Lag 7)	No remaining ARCH effects	0.7454

As reported in Table 9, the weighted Ljung-Box test on squared standardized residuals fails to reject the null hypothesis at conventional significance levels (p-value = 0.5734). Similarly, the ARCH-LM test confirms that there are no remaining ARCH effects (p-value = 0.7454). We can confidently conclude that the GJR-GARCH(1,1) model with Student-t errors adequately captures the volatility clustering in the FTSE All Share Index.

7 Conclusion

This paper investigated the volatility dynamics of the FTSE All Share Index using daily return data from 2010 to 2023. Our empirical strategy began with an inspection of basic statistical properties, revealing that the index returns are non-normal, fat-tailed, and exhibit distinct volatility clustering. Formal ARCH-LM tests strongly rejected the assumption of homoskedasticity, motivating the use of conditional volatility models.

By estimating GARCH(1,1) and TARCH/GJR-GARCH(1,1) models, we established two main findings. First, volatility in the broad UK equity market is highly persistent;

periods of turbulence decay slowly over time. Second, we found strong evidence of asymmetric volatility, commonly known as the leverage effect. Negative market shocks drive up uncertainty significantly more than positive shocks of an equivalent scale.

Model comparisons based on information criteria and log-likelihood favored the asymmetric models over the baseline symmetric GARCH. Final diagnostic checks verified that our preferred specification successfully removed remaining autocorrelation and ARCH effects. Ultimately, these results underscore that investors and risk managers dealing with UK equities must account for both long-lasting volatility regimes and the disproportionate market reactions triggered by bad news.

References

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Brooks, C. (2019). *Introductory Econometrics for Finance*. Cambridge University Press.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

Fidrmuc, J., and Horvath, R. (2008). Volatility of exchange rates in selected new EU members: Evidence from daily data. *Economic Systems*, 32(2), 103–118.

Glosten, L. R., Jagannathan, R., and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370.

Zakoian, J.-M. (1994). Threshold heteroskedastic models. *Journal of Economic Dynamics and Control*, 18(5), 931–955.

A Additional Tables and Figures

To ensure our findings regarding the leverage effect are not merely an artifact of the specific threshold mechanics in the GJR-GARCH model, we estimate an Exponential GARCH (EGARCH) model as a robustness check.

Table 10: EGARCH(1,1) Results (Student-t errors)

Parameter	Estimate	Robust Standard Error	p-value
ω	-0.0127	0.0050	0.0113
α_1	-0.1666	0.0210	0.0000
β_1	0.9654	0.0116	0.0000
γ_1 (Asymmetry)	0.1535	0.0271	0.0000

Consistent with our main results, the EGARCH specification confirms the asymmetric nature of volatility. In the `rugarch` parameterization, the sign effect parameter (γ_1) is estimated at 0.1535 and the size effect parameter (α_1) is negative (-0.1666), both being highly statistically significant (p-value = 0.0000).

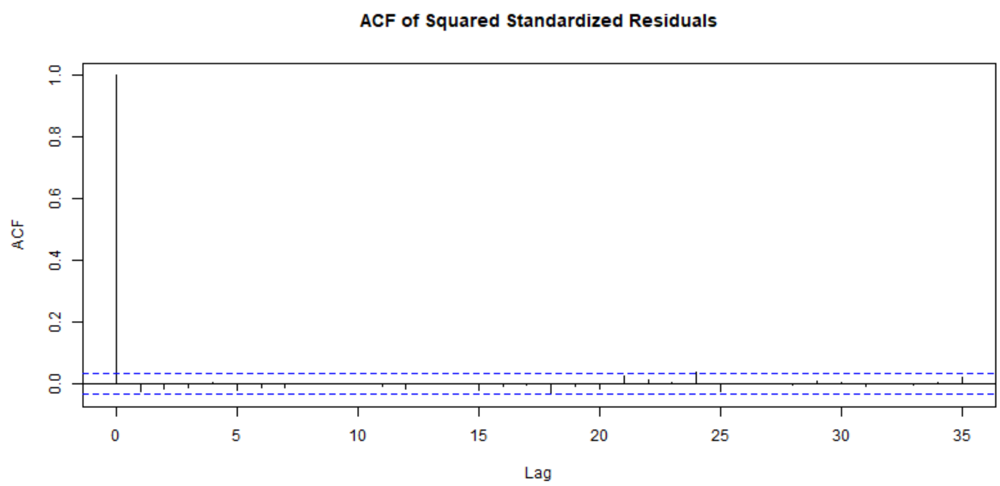

Figure 7: ACF of squared standardized residuals (GJR-GARCH model)

The ACF of squared standardized residuals in Figure 7 visually confirms that no significant autocorrelation remains in the variance of the residuals, proving that the GJR-GARCH model is well-specified.